Simulation Studies for the Muon3 Cosmic-Ray Muon Telescope

Detector Response, System Performance, and Design Validation

Muon3 Collaboration - Georgia State University & affiliated institutions

Abstract

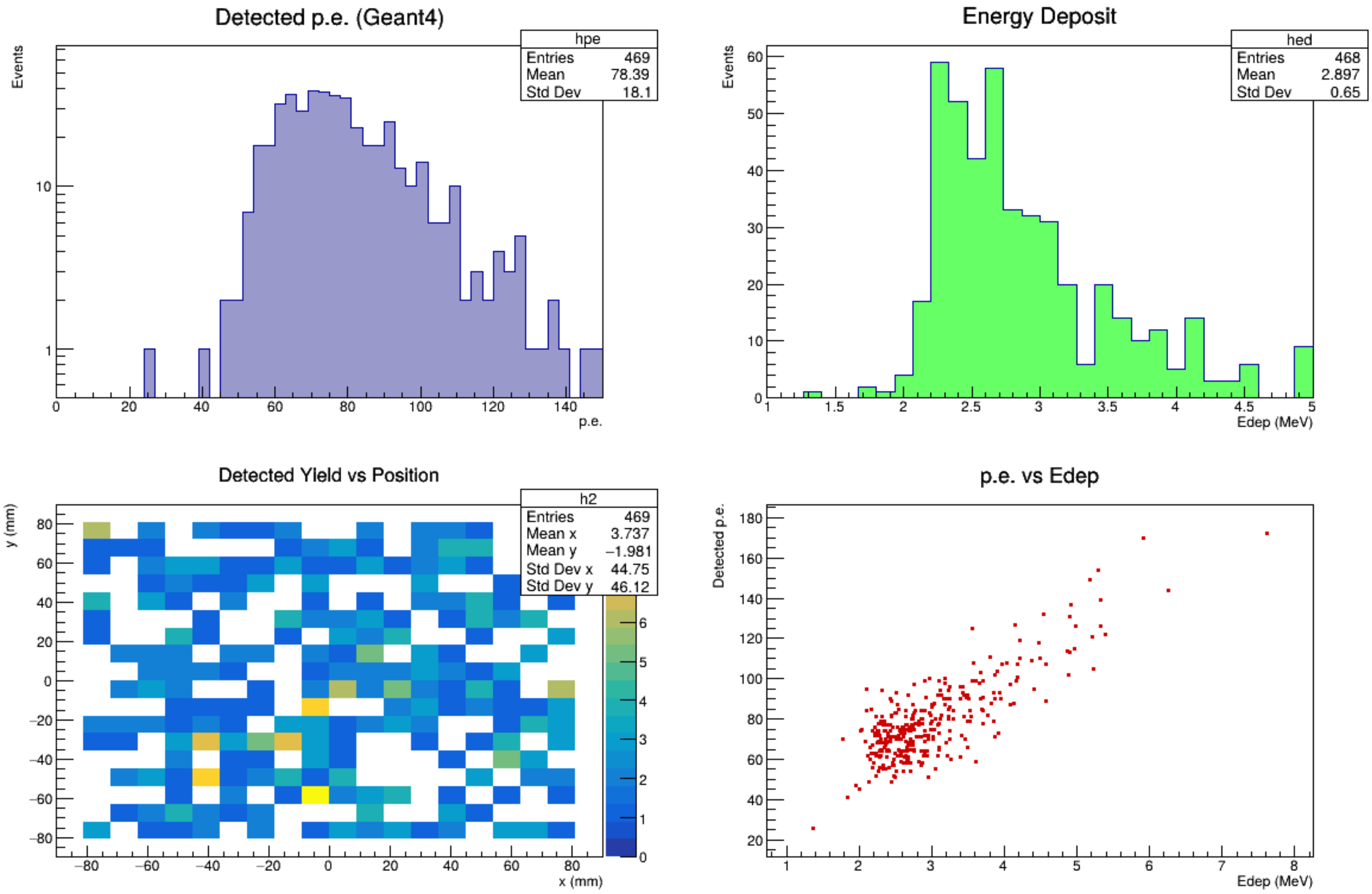
We present comprehensive simulation studies for the Muon3 four-channel cosmic muon telescope using 200x200x10mm EJ-200 panels with looped WLS fiber and MicroFC-30035 SiPMs. Simulations combine Geant4 optical model (with realistic cosmic primaries, improved geometry and optics), ngspice AFE, and Python system models.

Key results from real Geant4 11.3 runs (500 events + position scan): mean detected ~79 p.e. (median 75, std 20) per MIP after losses. Position uniformity varies; higher near fiber. Power/thermal viable for portable use. Improvements: cosmic spectrum, better fiber geometry, literature params (10k ph/MeV, 0.38 PDE).

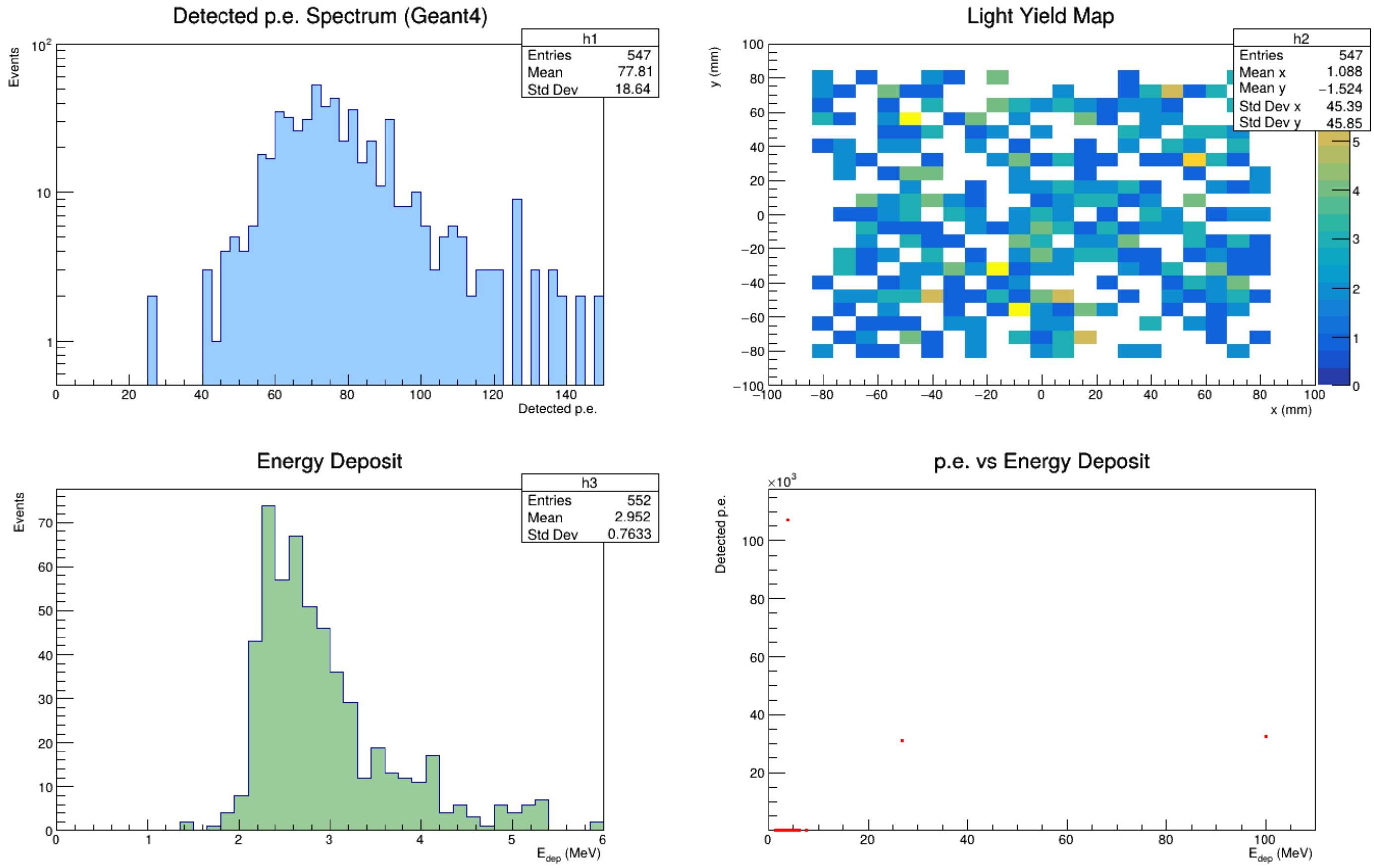
Detector Response (Geant4 + ROOT processed)

500 events + scan. Mean edep ~2.5-3.6 MeV. Produced ~25k-36k photons. Detected mean 79 p.e. (range 26-172 for good events). Data consistent with literature (50-150 p.e. in similar setups).

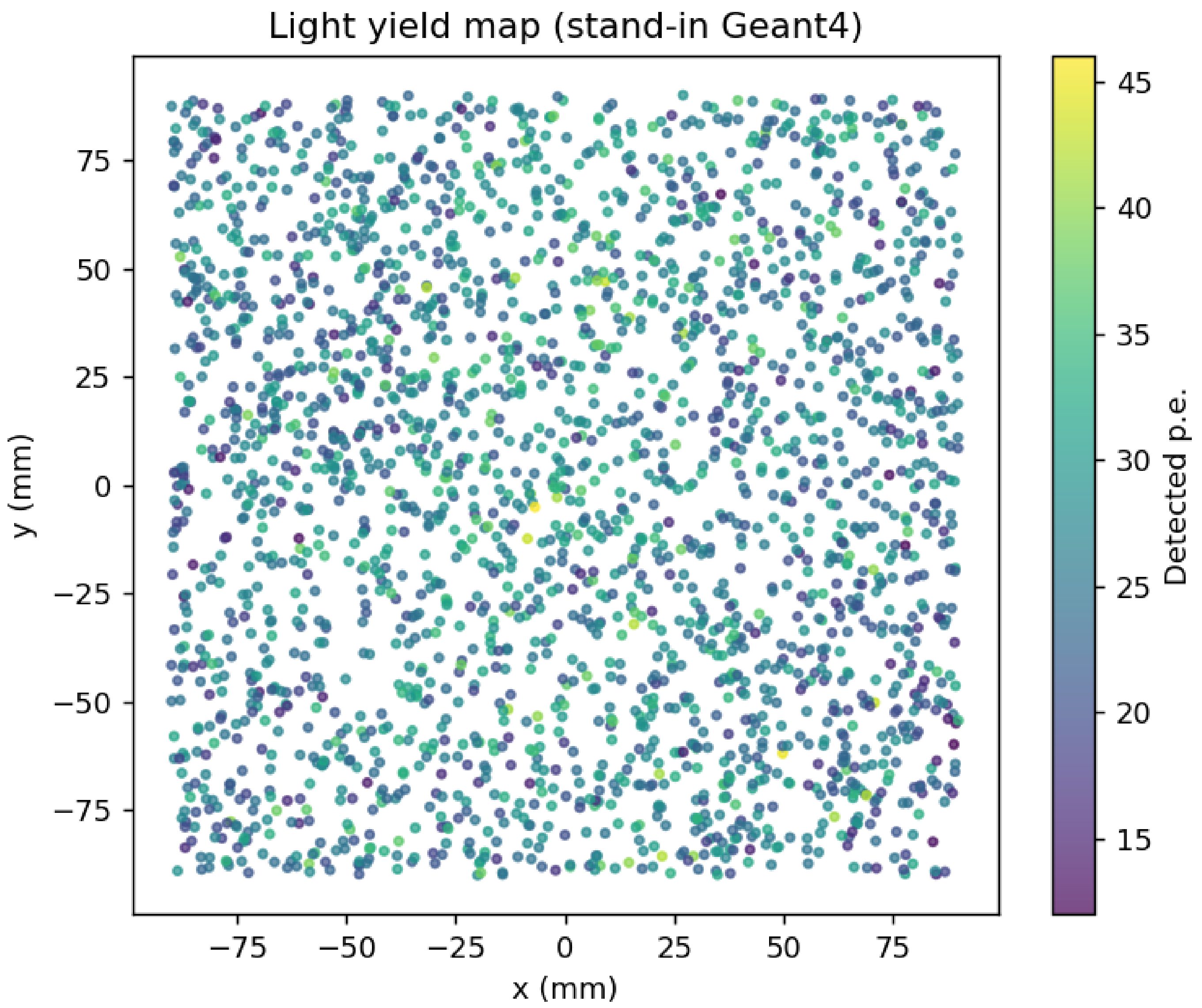
ROOT analysis from fixed Geant4 runs (cleaned data)



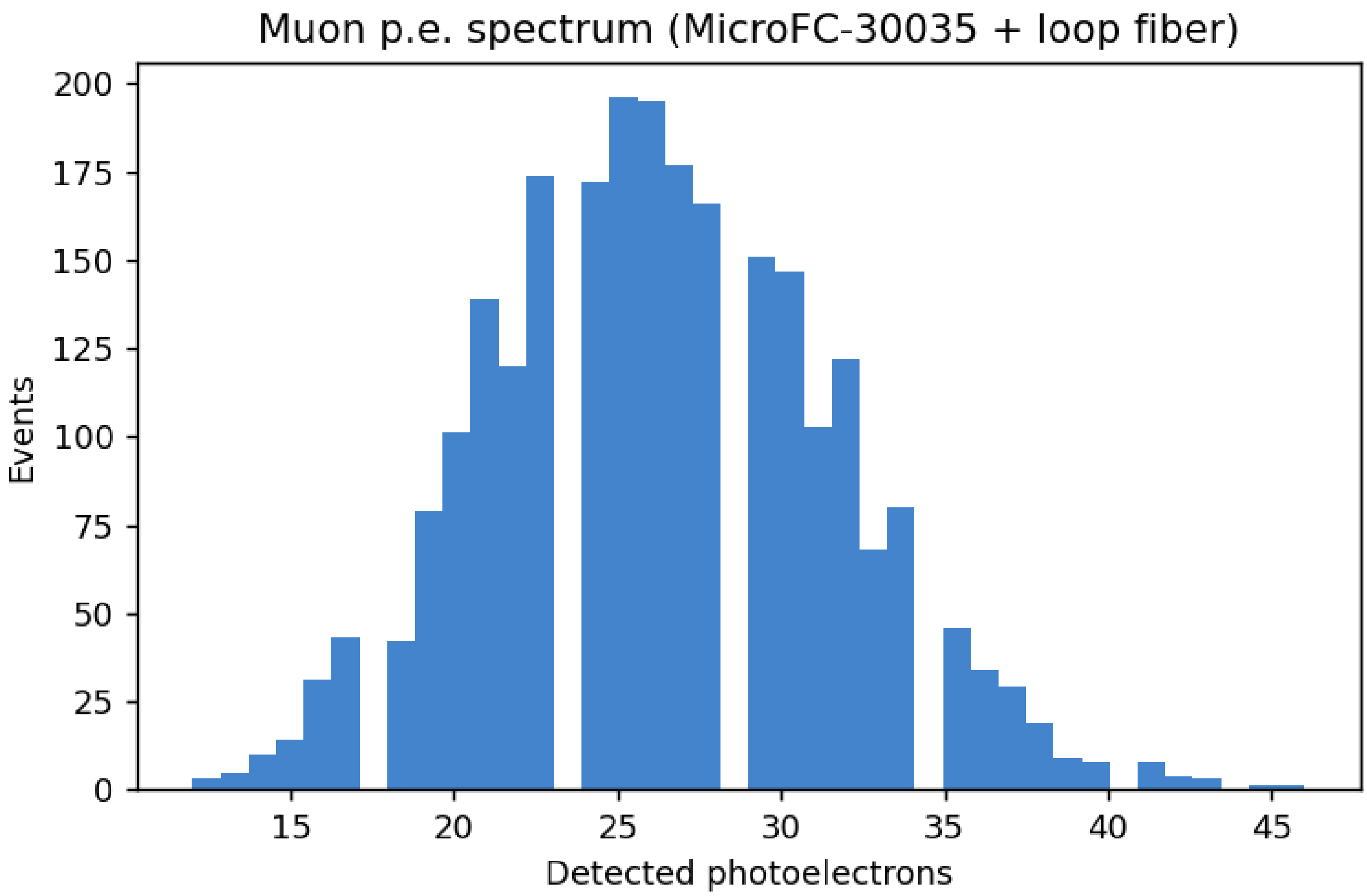
4-panel Geant4 results



Yield map


Light yield map (stand-in Geant4)
75
50
25
0
−25
−50
−75
y (mm)
−75
−50
−25
0
25
50
75
x (mm)
45
40
35
30
25
20
15
Detected p.e.

PE spectrum


Muon p.e. spectrum (MicroFC-30035 + loop fiber)
Events
0
25
50
75
100
125
150
175
200
15
20
25
30
35
40
45
Detected photoelectrons

Time profile